Syllabus Year: 2024-2025

CHRIST (Deemed to be University), Bengaluru - 560029

End Semester Examinations - November 2025

Bachelor of Computer Applications SEMESTER - I

Course Code: BCA103-1 **Max Marks: 50**

Course Name: PRINCIPLES OF SOFTWARE DEVELOPMENT - I **Duration: 02:00**

General Instructions:

- All rough work should be done in the answer script. Do not write or scribble in the question paper except your register number.
- Verify the Course code / course title and number of pages of questions in the question paper.
- Make sure your mobile phone is switched off and placed at the designated place.
- Malpractices will be viewed very seriously.
- Answers should be written on both sides of the paper in the answer book. No sheets should be detached from the answer book.
- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer book.

Course Outcome: Students will be able to:

CO1: Design and Develop ethical software product.

CO2: Apply terms, concepts, and tools of relational database management systems.

CO3: Construct database projects using DDL and DML queries.

Section A **2*5=10**

1)	Analyze the differences between entity and entity set in the context of relational database management systems. Provide examples to support your explanation.	**2**	(CO2, L4)
2)	Examine and differentiate between the concept of an algorithm and a data structure, highlighting their significance in software product development.	**2**	(CO1, L4)
3)	Examine and differentiate between the ethical considerations during requirement analysis and quality assurance phases in software development.	**2**	(CO1, L4)
4)	Analyze the differences between DDL and DML queries in constructing database projects.	**2**	(CO3, L4)
5)	Analyze and differentiate between the concepts of normalization and denormalization in database management systems.	**2**	(CO2, L4)

Section B **5*4=20**

6)	Construct a relational database model for a Library Management System using DDL queries, including at least three tables. Then, demonstrate how to create a view that displays all books borrowed by a specific member. Explain the advantages of using views in this context.	**5**	(CO3, L3)
7)	Describe the process of requirement elicitation and discuss its importance in software development by illustrating an example where inadequate elicitation led to a project failure.	**5**	(CO2, L2)
8) a)	Demonstrate the differences between an entity and an entity set by constructing examples from a real-world scenario such as a university management system.	**5**	(CO2, L3)
	OR		
b)	Construct an example scenario where you would need to elicit requirements for a new system and explain the potential quality issues that could arise if these requirements are not accurately captured.	**5**	(CO2, L3)

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9)	Analyze how database quality assurance (QA) practices can impact software development projects. Provide examples to illustrate your points.	**5**	(CO2, L4)

Section C **10*2=20**

10)	Construct a scenario where an ethical dilemma arises during requirement elicitation for a new social media platform. Identify and explain the potential ethical issues that may emerge, and propose strategies to mitigate these issues while ensuring the development of a quality software product.	**10**	(CO1, L3)
11) a)	Discuss the various layers involved in software development and explain their importance in the overall process of designing ethical software products.	**10**	(CO1, L2)
	OR		
b)	Discuss the key differences between an entity and an attribute within a relational database model, using specific examples to illustrate your points. Additionally, describe how views can be utilized effectively in managing data access and retrieval, and provide step-by-step instructions for creating, inserting into, and deleting from a view.	**10**	(CO1, L2)

Revised Bloom's Taxonomy (RBT) Levels

L1 - Remember
L2 - Understand
L3 - Apply
L4 - Analyse
L5 - Evaluate
L6 - Create

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